

Python Programming Tips

1. Use List Comprehensions

List comprehensions are faster and more readable than traditional for loops. Example: `squares = [x**2 for x in range(10)]`

2. Context Managers

Use 'with' statements for file operations. They automatically close resources even if exceptions occur. Example: `with open('file.txt') as f: content = f.read()`

3. f-strings for Formatting

Python 3.6+ f-strings are the fastest and most readable way to format strings. Example: `f'Hello, {name}!'`

4. Type Hints

Type hints improve code readability and enable IDE autocomplete. Example: `def add(a: int, b: int) -> int: return a + b`

5. Virtual Environments

Always use venv to isolate project dependencies. Commands: `python -m venv venv`, then activate with `venv/Scripts/activate`

6. Enumerate Instead of range(len)

Use `enumerate()` to get both index and value in a loop. Example: `for i, item in enumerate(items): print(f'{i}: {item}')`

7. Defaultdict for Grouping

`collections.defaultdict` simplifies grouping operations significantly. Example: `from collections import defaultdict; groups = defaultdict(list)`

8. Dataclasses for Data

Python 3.7+ `@dataclass` decorator auto-generates `__init__`, `__repr__` and more. This reduces boilerplate code significantly in data-oriented classes.